

A hybrid Black-Scholes evaluator: recovering speed with bounded synthetic accuracy evidence

Khamit Kadyrbekov and Daniyal Kadirbekov Independent research, Tashkent, Uzbekistan Working note
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Abstract

The published v2 prototype evaluates a positive-payoff representation of the Black-Scholes-Merton price by quadrature. It avoids problematic subtraction in extreme synthetic examples, but the detailed Python API is slow. We replace quadrature in selected regions by an exact-input ATM identity, a small-volatility moment series, an asymptotic difference series, or a conditioned direct CDF evaluation. The unchanged v2 integrator remains a fallback. On 480 new locally frozen exact-binary64 input cases, both v2 and v3 pass all predeclared mixed price/log tolerances. Median batch time falls from 90.143 to 19.668 microseconds per call, a 4.583-fold speedup on this machine and synthetic mixture. On 426 nonzero rounded reference prices, v3 is closer to the reference than v2 in 76 cases, farther in 99, and tied in 251. Thus speed is recovered without breaking the declared tolerances, not without every change in numerical error. Analytic Taylor remainder arguments concern exact-arithmetic truncation; total floating error is not certified. No new market model, global superiority, historical novelty, or economic return is established.

Substantial AI assistance (Codex) was used for derivation, implementation, experiment design, execution, analysis, and drafting. Human independent reproduction has not yet occurred. Khamit initiated and directed the research; Daniyal is coauthor and the designated lead for continuing software work. This does not assert that Daniyal independently implemented or validated this frozen prototype. Historical-mathematician AI roles are not human consultants.

1. Contract and positive-payoff representation

All six inputs S, K, T, r, q, σ are interpreted as their exact binary64 values by the high-precision reference. Let $X = S \exp(-qT)$, $Y = K \exp(-rT)$, $m = \log(S/K) + (r-q)T$, and $v = \sigma \sqrt{T} > 0$. Let $L = X$ if $m \leq 0$ (OTM call), or $L = Y$ if $m > 0$ (OTM put), and $z = -|m|/v + v/2$. The positive OTM price is

$$P = L \phi(z) B(z, v),$$
$$B(z, v) = \int_0^\infty \exp(zu - u^2/2) [1 - \exp(-vu)] du.$$

The opposite option follows from put-call parity. v2 and v3 use the same $\log 1p/\text{frexp}$ moneyness, split-exponent monetary reconstruction and parity calculation. Factoring monetary scale before reconstruction avoids prematurely rounding a normalized price to zero. The candidate uses ordinary Python/libm binary64 only, with no oracle or multiprecision calls.

2. Small-volatility moment expansion

Define $I_n(z) = \int_0^\infty u^n \exp(zu - u^2/2) du$. Then

$$B(z, v)/v = \sum_{n=1}^N [(-v)^{n-1}/n!] I_n(z) + R_N,$$
$$|R_N| \leq v^N I_{N+1}(z)/(N+1)!.$$

This finite expansion and bound follow by integrating the pointwise Taylor remainder of $\exp(-vu)$ against a nonnegative weight. No exchange with an unbounded infinite series is required. Integration by parts yields

$$I_0 = \sqrt{(\pi/2)} \exp(z^2/2) \operatorname{erfc}(-z/\sqrt{2}),$$

$$I_1 = 1 + zI_0, \quad I_n = (n-1)I_{n-2} + zI_{n-1}.$$

The implementation restricts $-1 \leq z \leq 1/16$ and $0 < v \leq 1/8$ to avoid the severe cancellation of this forward recurrence at large negative z . It stops when the first omitted term is $\leq$ one binary64 epsilon times the partial sum, with at most 17 included moments. Failed positivity or convergence uses $v2$. The computed stopping estimate does not include recurrence or libm rounding.

3. Tail expansion of the difference

Write $a = -z > 0$ and $\rho = v/a$. Expanding $\exp(-u^2/2)$ inside the positive integral, then integrating each finite term against $\exp(-au)(1 - \exp(-vu))$, gives

$$B(-a, v) = (v/a^2) \left[\sum_{n=0}^{N-1} (-1)^n (2n-1)!! a^{-2n} H_{2n+1}(\rho) + E_N \right],$$

$$H_p(\rho) = [1 - (1+\rho)^{-p}]/\rho.$$

where $(-1)!! = 1$ and $H_p(0) = p$. The finite Taylor remainder gives

$$|E_N| \leq (2N-1)!! a^{-2N} H_{2N+1}(\rho).$$

This is a bound for the **difference** under a positive weight; subtracting two separately approximated Mills ratios is unnecessary. Evaluate H_p by $-\expm1(-p \log 1p(\rho))/\rho$, with its limiting value when ρ rounds to zero. The implementation tries this only for $a \geq 12$, uses at most 64 omitted-term checks, and abandons a nondecreasing term sequence before convergence. v/a^2 remains factored until monetary reconstruction. General erfc asymptotics and their remainder behavior are standard; see [NIST DLMF §7.12](#). The particular implementation and experiment are not claimed historically new.

4. ATM, conditioned direct path, and fallback

For the exact-input case $S=K$ and $r=q$,

$$P = L \operatorname{erf}(v/(2\sqrt{2})).$$

For $v < 1e-4$ use $v/\sqrt{2\pi}$ times $(1 - v^2/24 + v^4/640)$; the next normalized term has magnitude $v^6/21504$. Keeping v factored handles the smallest positive subnormal volatility when a large monetary scale makes the final price normal. This extreme is a separate unit fixture, not part of the 480-case confirmation. Rounded $m=0$ alone is not sufficient to enter the exact-input ATM branch.

Otherwise the direct normalized price is $\Phi(z) - \exp(|m|)\Phi(z-v)$. A heuristic screen combines subtraction condition, squared normal arguments and $|m|$. It requires arguments in $[-26, 26]$, $|m| \leq 600$, and score $\leq 1e-13$. This is not a libm error certificate. Every region not accepted uses the unchanged $v2$ integral; $v3$ still explicitly rejects unsupported ranges. Greeks, branchwise derivatives and implied-volatility inversion are outside this experiment.

5. Experimental integrity and accuracy

All 604 $v1/v2$ inputs were development data. They passed the existing mixed price/log/side gates using saved 35-digit reference strings. Twenty-eight unit test methods passed, including independent high-precision integral checks of the series, seams, scaling, parity, monotonicity and reciprocal symmetry.

Before the first new outputs, the candidate, all local lab modules/tests, protocol, timing code, native sources/binary, dependency hashes and 480 new inputs were frozen locally at 2026-09-26 07:53:38 UTC. Candidate SHA-256: 4bf8c51ca33082665b9c6902bb2a114604d937b6dd0ed67e0c17ece46b4547bc. This improves the earlier incomplete pipeline freeze; it remains local evidence, not an externally preregistered or

blinded experiment. Source and candidate hashes remained unchanged after outputs. The corpus contains 96 small-v, 80 tail, 80 regular-with-carry, 96 switching-neighborhood, 80 monetary-range and 48 exact-ATM cases. Thirteen nextafter steps removed input duplicates with old data; prices were not used in corpus generation.

The unchanged erfc oracle converged at 100/180 digits in every case; none needed the planned 260-digit retry. All 12 fixed supplementary integral checks passed at 100 digits. They use another numerical integrator but a mathematical representation related to the candidate; they are not an independent mathematical formulation. All 80 carry-adaptor oracle checks also converged.

Both v2 and v3 pass all 480 gates: finite nonnegative call/put, correct OTM side, no unexpected zero, price error $\leq \max(1e-10 \text{ times reference, one ULP of rounded reference})$, and log error $\leq \max(2e-11, 16 \text{ ULP of rounded reference log})$. These are mixed practical tolerances, not universal correct rounding.

Quantity	Result for v3
Normal / subnormal / rounds-to-zero references	375 / 51 / 54
Maximum relative error, normal prices only	3.1224e-13
Maximum absolute log error, all 480	6.0008e-13
Rounded-reference matches, 426 nonzero prices	134
Rounded-reference matches, 51 subnormals	40
Exact ATM / small-v / tail / direct / fallback counts	48 / 125 / 176 / 63 / 68

Very small subnormals require a separate interpretation: in case v3-0353, the exact price is about $2.7081e-324$ and the correctly rounded result is the smallest positive binary64 number, about $4.9407e-324$. Its relative error is 82.44% despite correct rounding. Do not advertise $3.1224e-13$ for all nonzero prices. Similarly, v3-0455 has log error about $2.0e-17$ but $1.58e12$ log-ULPs because the log is near zero; do not advertise uniformly small log-ULP error.

On the 426 nonzero rounded prices, smaller exact absolute error counts are:

Comparator	v3 closer	v3 farther	Equal error
Own original v0	398	12	16
Frozen integral v2	76	99	251
Historical native Jaeckel	329	61	36

The historical comparator is mirror commit [83ae882df8e19323798c7ebfb8898f94d2d92ade](#), not verified latest official upstream. Its comparison includes the forward/discount adaptor, whose distortion is recorded per carry case. These results say nothing about a newer version or the frequency of such inputs in trading. v0 produces 50 unexpected zeros, the historical adaptor/native combination 56, and v2/v3 none on these 426 prices. Both v2 and v3 retain finite logs when monetary prices round to zero.

6. Timing and the explicit compromise

Timing used the detailed Python APIs, one sequential CPU worker, no GPU, one warm-up and seven alternating-order batches with five loops each. The table reports ratios of median batch times, not medians of per-call ratios.

Fixed stratum	Cases	v2 microseconds/call	v3 microseconds/call	v2/v3
All new inputs	480	90.143	19.668	4.583

Fixed stratum	Cases	v2 microseconds/call	v3 microseconds/call	v2/v3
Exact ATM	48	75.445	3.238	23.304
Small volatility	96	93.113	6.174	15.082
Negative tail	80	92.562	8.019	11.543
Monetary range	80	92.998	8.089	11.497
Regular with carry	80	84.384	27.526	3.066
Switching neighborhoods	96	93.469	54.087	1.728
Fallback subset, selected by v3 path	68	94.025	96.706	0.972

The fallback subset is about 2.85% slower because the selection work precedes integration; it is not an additional independent stratum. Its v3 price and log outputs are identical to v2. Across all inputs v0 takes 5.307 microseconds: v3 remains 3.706 times slower than that less accurate baseline. No C++ speed comparison or production-throughput claim is made. The predeclared criterion (all gates plus >2x all-case speedup over v2) is met. The main result is a measured accuracy/speed tradeoff, not that v3 beats every comparator everywhere.

7. Limits, reproduction, and next work

Converged multiprecision and exact-arithmetic remainder formulas are not a certified floating-point error proof. Binary64 input transformation, libm, recurrence rounding, subnormal rounding and parity reconstruction remain separate error sources. The stress mixture overweights extreme ranges. There is no full-domain test, independent human validation, model calibration, market-loss estimate, Greek accuracy assessment, or implied-volatility study.

Run from the project root with Python and mpmath 1.3.0, one numerical thread:

```
python3 -m unittest discover -s tests -v
python3 reproduce_v3.py --output /tmp/black-scholes-v3-reproduction.json
```

The local freeze pins this machine's native binary and shared one-worker wrapper. The portable harness was added **after** the original result. It validates original source/input and mpmath hashes, compiles the native comparator sequentially in a temporary directory, and substitutes only its location and the source verifier. Its report explicitly labels the new build/environment; it does not rewrite the original freeze or claim external independent validation. The native C++ wrapper hash is pinned transitively by the frozen native-build receipt and is checked explicitly by this replay harness. Native sources and license notices are preserved. All raw rows, timing batches and pipeline hashes are in evidence/benchmark-v3.json, evidence/python-timing-v3.json, and evidence/freeze-v3.json; the exact protocol is EXPERIMENT_V3.md.

The next useful investigation is Greek continuity/accuracy at branch seams and robust conditioned IV inversion on a newly frozen corpus. A separate study could replace heuristic direct-path admission with verified error control. Existing v2 and v3 evidence must remain unchanged.

Sources and predecessor

- [Published v2 working paper and evidence.](#)
- [NIST DLMP §7.12: erfc asymptotics.](#)
- Pinned historical Jaeckel/Cody source and provenance: vendor/ and JACKEL_SOURCE_REVIEW_2026-09-26.md. Latest official Jaeckel PDF retrieval returned HTTP 502 during this continuation; no latest-version claim is made.